

§11.3 - The Integral Test and Estimates of Sums

Theorem #1

Let f be a positive, decreasing, continuous function on $[1, \infty)$ and define (a_n) by

$$a_n = f(n).$$

Then the following are equivalent:

1. The improper integral

$$\int_1^{\infty} f(x) \, dx$$

converges

2. The series

$$\sum a_n$$

converges.

Example #1

Determine if

$$\sum \frac{1}{n^2}$$

converges.

Example #2

Determine if

$$\sum \frac{1}{n}$$

converges.

Definition #1 (p -series)

Let $p \in \mathbf{R}$.

The sum

$$\sum \frac{1}{n^p}$$

is a p -series.

Theorem #2

The p -series

$$\sum \frac{1}{n^p}$$

converges if and only if $p > 1$.

Definition #2 (Harmonic Series)

We call

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

the **harmonic series**.

Theorem #3 (Integral Test Error Estimate)

Assume f and (a_n) satisfy the hypotheses of the Integral Test and

$$S = \sum_{k=1}^{\infty} a_k$$

converges.

Then

$$\int_{n+1}^{\infty} f(x) \, dx < S - S_n < \int_n^{\infty} f(x) \, dx$$

where S_n is the n th partial sum of S .

Remark

The following are equivalent:

1.

$$\int_{n+1}^{\infty} f(x) \, dx < S - S_n < \int_n^{\infty} f(x) \, dx$$

2.

$$S_n + \int_{n+1}^{\infty} f(x) \, dx < S < S_n + \int_n^{\infty} f(x) \, dx$$

Example #3

Let

$$S = \sum_{k=1}^{\infty} \frac{1}{k^2}.$$

1. Approximate S using S_{10} and estimate the error.
2. How many terms are required to ensure that the approximation by S_n has an error less than 10^{-2} ?

Example #4

Let

$$S = \sum_{k=1}^{\infty} \frac{1}{k^2}.$$

Use L_{10} to obtain a better approximation of S and estimate the error.

Remark

With some work, we can show

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

Example #5

Determine if

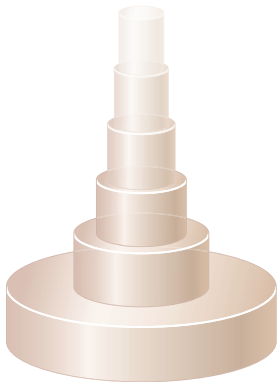
$$\sum_{k=1}^{\infty} \frac{\arctan(k)}{k^2 + 1}$$

converges.

Example #6

Consider a wedding cake of infinite height, each layer of which is a right circular cylinder of height 1.

The bottom layer of the cake has a radius of 1, the second layer has a radius of $1/2$, the third layer has a radius of $1/3$, and the k th layer has a radius of $1/k$.



1. Find the area of the lateral (vertical) sides of the wedding cake.
2. Determine the volume of the cake.